

Count Each Idea Once: Building Robust Composites from Correlated Alpha Signals

Berkeley MFE Industry Project, sponsored by Ultramarin

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Abstract

A quantitative research team rarely holds one definition of a factor. It holds a cluster of correlated variants of the same idea, and those variants must somehow become one signal. This report asks how such a cluster should be combined into a single composite, and whether anything beats the benchmark of ranking each variant and averaging. On four anonymized feature classes provided by Ultramarin (roughly 1,500–3,100 in-sample trading days each, ~ 390 –480 names per day), I evaluate composition methods ranging from equal weighting through supervised regressions, partial least squares, gradient-boosted trees, redundancy-aware clustering, and a random-effects estimator borrowed from clinical meta-analysis, under one fixed evaluation framework: daily rank information coefficients with Newey–West significance, signal decay, factor turnover, and purged cross-validation, followed by a final frozen test on a blind hold-out of roughly 2,180 unseen trading days per class. The resulting findings led to a robust methodology supported by evidence. *Count each idea once*: cluster near-duplicate variants into blocks, average within each block, weight the blocks equally, add learned per-feature transforms only where in-sample evidence clearly supports them, and freeze every parameter. No method in the study, including this one, delivers more raw predictive accuracy than the benchmark. What the selected composite improves is everything around the accuracy: it matches the benchmark on every hold-out class while carrying the smallest worst-case loss when the feature set is perturbed, and where the two differ it decays more slowly and trades at 35–40% lower turnover. In short, the methodology keeps only what the evidence supports, and documents why every rejected alternative was left out.

1 Introduction

Feature definitions in finance are arbitrary in a specific, well-known way. The literature settles on one definition of momentum, say twelve-month return skipping the most recent month, but a live research process accumulates dozens: different lookbacks, different normalizations, risk-adjusted and residualized flavors, each a noisy measurement of the same latent signal. This report investigates how a large set of highly correlated features, all manifestations of one signal theme, can be combined into a single robust composite, and whether any weighting scheme can beat the simplest one.

That simplest scheme is the benchmark throughout this report: rank each variant cross-sectionally, then average the ranks with equal weights. Writing $r_{j,t}$ for variant j 's cross-sectional rank on day t and s_j for the sign of its training-window IC, the benchmark composite on day t is

$$F_t = \frac{1}{N} \sum_{j=1}^N s_j r_{j,t}, \quad (1)$$

where the sign flips any variant whose training relationship with the target is negative (on percentile ranks a reflected variant differs from $-r_{j,t}$ only by a constant, which no ranking sees). We refer to it as the *benchmark* or the rank-and-average composite. It is harder to beat than it looks, but it has one structural flaw: it weights definitions, not ideas. For example, if eight of ten variants are re-implementations of “cumulative return,” the average is really that one idea, eight times over, with the other two ideas diluted to noise. The objective is therefore a systematic, statistically sound way to weight variants by uniqueness and information content rather than by how many near-duplicates happen to land in the cluster.

The textbook dimensionality-reduction answer, PCA, is unfortunately not the answer here. PCA extracts the linear combination with maximum variance, while our objective is maximum correlation with residualized forward returns, a target the PCA objective never sees. On redundant clusters the two disagree in exactly the wrong direction: the near-duplicates share most of the common variance, so the first principal component concentrates on the most-repeated definition, which reproduces the benchmark's flaw in a more roundabout form. The cost is measurable. On the one class in our data with several distinct sub-themes, a frozen first-principal-component composite keeps about a third of the benchmark's out-of-sample IC.

To answer the question systematically, I built a single fixed evaluation framework (daily rank IC, Newey–West t -statistics, signal decay, factor turnover, purged k -fold cross-validation) and pushed a sequence of composition methods of increasing complexity through it: equal-weight and IC-weighted averages, GLS-style “uniqueness” regressions, penalized regressions, partial least squares, gradient-boosted trees, cluster-then-weight composites, and a tuning-free random-effects weighting borrowed from clinical meta-analysis. Everything that survived in-sample scrutiny was then frozen and scored once on a blind hold-out: roughly 2,180 trading days per class that arrived after all in-sample decisions were locked. Two statistical checks close the study: a family-wise multiplicity correction across all 56 method-versus-benchmark comparisons the project ever ran, and estimator-robustness checks on the one data-driven decision the final method retains.

From the outset, the project favored transparent reporting over good-looking results, and that mandate shaped it more than any single method did. Three findings define the story.

1. Each cluster is worth what it is worth, and no weighting scheme extracts more. Every class has a hard IC ceiling (0.031 for the strongest, 0.015 for the weakest) that equal weighting essentially attains and that trees, PLS, GLS, and tuned penalties cannot move. After multiplicity correction, no method in the study beats the benchmark on raw IC (Section 8).

2. Deduplication is the one idea that survives every test. Clustering the variants, collapsing each redundancy block to its sign-aligned average, and weighting blocks equally matches the benchmark everywhere while never being the casualty: it has the shallowest worst case across 80 random feature-subset perturbations, no blow-up on any hold-out class, and the study’s only nominally significant win against the benchmark (on the multi-factor class, where the benchmark’s duplication bias actually binds).
3. The selected composite’s edge is signal quality, not the level of its IC. Every class has its own ceiling, and the selected composite sits at or within noise of all four of them, which no single rival method manages: trees collapse off class 1, regression fits invert on class 3, and PCA collapses on class 4. Where a class has learnable shape structure the composite adds it, and it then holds its IC at longer holding horizons at materially lower factor turnover, so it trades more cheaply than the benchmark wherever the two differ at all.

2 Related work

Combining correlated noisy predictors of one quantity is the oldest problem in the forecasting literature (Bates and Granger, 1969; Clemen, 1989; Timmermann, 2006). Its most robust empirical finding, the “forecast combination puzzle,” is that estimated “optimal” weights routinely lose to equal weights out of sample, because the weights themselves are estimated with error comparable to the differences they chase (Stock and Watson, 2004; Smith and Wallis, 2009). The portfolio-choice analogue is DeMiguel et al. (2009): no estimated allocation consistently beats $1/N$. Our study effectively replays this literature inside a single alpha signal, and reaches the same resolution the literature does: shrink toward equal weights, and deviate only on strong evidence. What we add is the deduplication observation. On redundant clusters the right unit for “equal” is the idea (the redundancy block), not the raw variant, a distinction the classical combination literature does not need because its forecasters are not near-duplicates.

The mechanism behind the puzzle has a name and a remedy. It is the Stein phenomenon (James and Stein, 1961; Efron and Morris, 1977): when many means are estimated jointly, shrinking toward a common value dominates the raw estimates, and Ledoit–Wolf covariance shrinkage (Ledoit and Wolf, 2004) applies the same logic to covariance matrices. Both appear in our results in recognizable form. The “uniqueness” regression of Section 5.1 only works when its ridge penalty is large, that is, when it stops trusting the estimated covariance, and the random-effects estimator of Section 5.3 is exactly a James–Stein shrinkage of per-block ICs with a data-estimated intensity.

The cleanest off-the-shelf implementation of that shrinkage idea comes from an unexpected field. Pooling k correlated, noisy effect estimates is the core problem of research synthesis in medicine (Cochran, 1954; Borenstein et al., 2010), and the random-effects model of DerSimonian and Laird (1986) estimates the between-study heterogeneity τ^2 in closed form, interpolating between equal weighting ($\tau^2 = 0$) and precision weighting (τ^2 large). We import it as a tuning-free adaptive weighting across redundancy blocks, with alternative τ^2 estimators (Paule and Mandel, 1982; Harville, 1977; Veroniki et al., 2016) and the Hartung–Knapp small- k correction (Hartung and Knapp, 2001) as robustness checks in Section 8. To our knowledge the explicit mapping (feature variant as study, train-window IC as effect estimate, Newey–West standard error as within-study error, duplicated definitions as duplicate publications) is not standard in the signal-combination literature, and Section 5.3 spells it out.

Whatever the weighting scheme, the evaluation has to be trustworthy, and here we follow the standard institutional playbook: information coefficients and the breadth arithmetic of Grinold and Kahn (2000); HAC standard errors (Newey and West, 1987), because overlapping 21-day tar-

gets induce strong serial correlation in daily ICs; and purged k -fold cross-validation with embargo (López de Prado, 2018), because random splits leak overlapping labels. A research process that runs many backtests will find “significant” ones by construction, the multiplicity problem of Harvey et al. (2016). We implement their prescription literally in Section 8, with Holm (Holm, 1979), Benjamini–Hochberg (Benjamini and Hochberg, 1995), and the dependence-aware Westfall–Young max- T bootstrap (Westfall and Young, 1993) over every comparison the project ever ran.

Finally, the nonlinear probes use gradient boosting (Friedman, 2001; Chen and Guestrin, 2016; Ke et al., 2017). The finding that a depth-1 (stump) ensemble, exactly a generalized additive model (Hastie and Tibshirani, 1986), reproduces everything the deeper trees deliver places our results in a familiar tradition: on tabular financial data the win, when there is one, is per-feature nonlinearity, not interactions. The two-compartment decay decomposition mentioned in Section 6.5 borrows its functional form from pharmacokinetics (Gibaldi and Perrier, 1982).

3 Data and exploratory analysis

3.1 The panels

Ultramarin provided four anonymized *feature classes*, referred to as class 1 through class 4 throughout this report. Each class is one cluster of correlated feature variants $f_1, \dots, f_N$, delivered as within-day cross-sectional ranks on a panel of anonymized equity identifiers. Each f_j is, at least in principle, a representation of the same latent feature. Alongside the variants comes a single target: residualized forward returns. Dates are integer-indexed trading days, and identifiers are anonymized but consistent within a class. Table 1 summarizes the panels after preprocessing to complete cases, that is, dropping rows with a missing target or any missing variant. This removes under 0.4% of rows for classes 1 and 4 and about 12% for classes 2 and 3 in sample, and under 1% everywhere on the hold-out.

Table 1: The four feature classes. “PC1 var. share” is the fraction of within-class feature variance explained by the first principal component of the pooled panel correlation matrix of the variants (all days stacked, one correlation matrix per class, not a day-by-day PCA): a measure of how close the cluster is to one factor. The blind hold-out (Section 7) is the immediately following period for every class and was delivered only after all in-sample analysis was frozen.

	variants	in-sample		hold-out		names/day	PC1 var. share
		days	rows	days	rows		
class 1	16	3,130	1.51M	2,195	0.93M	~481	0.48
class 2	10	1,565	0.61M	2,195	0.93M	~390	0.51
class 3	6	1,565	0.61M	2,195	0.93M	~390	0.53
class 4	20	3,130	1.51M	2,195	0.93M	~481	0.30

3.2 The target

The target we model is a residualized cumulative forward return over approximately 21 trading days (one month), sampled every day. Because consecutive days’ targets share 20 of their 21 forward days, the target’s autocorrelation decays linearly to zero at lag 21 (Figure 1). Its cross-sectional dispersion, the standard deviation of the target across names on a given day, is about 6.6% and annualizes to a sensible ~23% only at the monthly horizon, consistent with a monthly return sampled daily. This overlap is the one property of the target that matters for everything

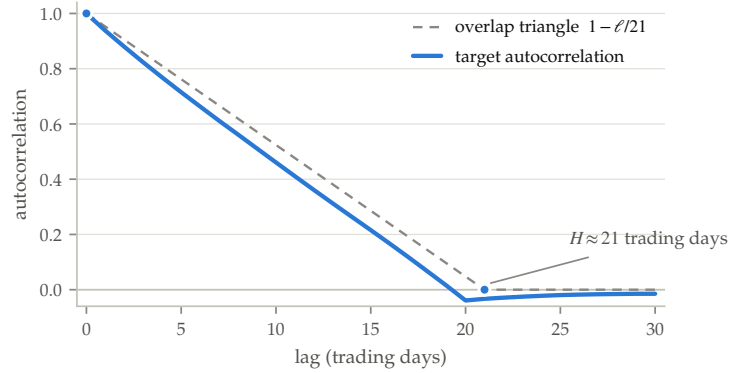


Figure 1: The returns being modeled. The target’s autocorrelation traces the overlap triangle of a ~ 21 -day cumulative forward return sampled daily; this one property sets the purge window and the HAC lag count for the entire study.

downstream: it fixes the 21-day purge-and-embargo window in cross-validation and the 25-lag width of every Newey–West standard error in the study.

3.3 Redundancy and orientation

Before any averaging is meaningful, all variants in a cluster must point the same way relative to the target. That requirement cannot be met by feature-to-feature correlations alone, which are themselves noisy and say nothing about which direction predicts. A minimal level of supervision is therefore unavoidable: at least one sign per cluster must come from the target, and in the composites below every variant is ultimately flipped by the sign of its training-window IC. What unsupervised structure can still tell us is how the variants relate to each other, and that is what this section examines.

Figure 2 shows each class’s correlation matrix three ways: raw, grouped by sign convention, and grouped by redundancy after *sign alignment*. Alignment reflects a variant ($x \mapsto 1 - x$ on percentile ranks) if and only if it loads negatively on the cluster’s first principal component, a convention fix that changes no correlation magnitude. The three views separate two things that are easy to conflate.

- **Orientation.** Class 1 looks like a mixed positive/negative grid, but flipping just 2 of its 16 variants collapses it into a single coherent positive block (mean off-diagonal correlation $0.21 \rightarrow 0.44$, negative-pair fraction $0.23 \rightarrow 0$). One theme, inconsistent sign conventions. Classes 2 and 3 arrive already coherent.
- **Real multi-factor structure.** Class 4 is an exception to the others: five variants get flagged, but flipping them does not clean the matrix (mean off-diagonal actually falls, $0.23 \rightarrow 0.15$). Its negative correlations are real disagreement between sub-themes, not reversed duplicates. This is the cluster where uniqueness-aware weighting should finally earn its keep, and the one where it eventually (nominally) does. It is also exactly the case the prerequisite above predicts: when a cluster is not one theme, a PC1-based sign convention cannot substitute for target-based signs, and it is the training-IC sign bits that resolve class 4.

Within-class redundancy is otherwise exactly as the project premise describes: solid blocks of 0.7–0.95 correlation, many variants being near-duplicate re-implementations, and this duplication is precisely what the benchmark overweights. Two further exploratory facts constrain what follows. First, within a class the variants’ $|IC|$ values are broadly similar. No definition dominates, so

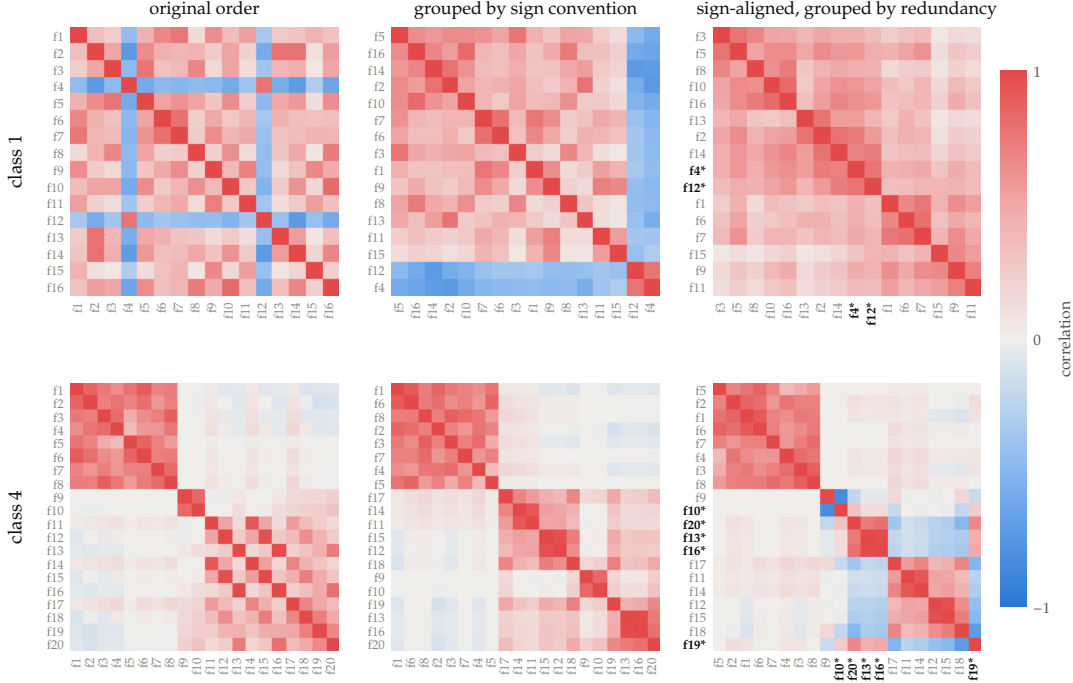


Figure 2: The two classes that carry the orientation story: correlation matrices raw (left), grouped by sign convention (middle), and sign-aligned then grouped by redundancy (right; * marks reflected variants). Class 1 (top) collapses to one coherent block after flipping two variants; class 4 (bottom) retains real sub-block structure that no per-feature sign fix can reconcile. Classes 2–3 arrive already coherent.

sophisticated weighting has little cross-sectional signal to exploit and must still pay its estimation noise. Second, class 1 illustrates why the target’s sign bit is irreducible: after alignment its coherent orientation is *anti-predictive*, with all 16 variants carrying negative IC. A purely unsupervised composite of this cluster is not just weaker, it points the wrong way.

4 Evaluation framework

Every method in this study is scored by a fixed set of metrics, defined before any comparison was run. On day t , let F_t be the cross-section of composite scores and R_t the matching realized targets. The day’s *information coefficient* is the Spearman rank correlation $IC_t = \text{corr}(\text{rank}(F_t), \text{rank}(R_t))$. The focus on ranks is a scope decision, not a claim of universality: this study evaluates signals as orderings, which is robust to fat-tailed returns, while portfolio constructions that use the distance between predictions lie outside its scope. Daily equity ICs are small (0.02–0.05 is a strong signal), and all inference runs on the time series $\{IC_t\}$ through the following metrics:

metric	definition	reads as
mean IC	$\overline{IC_t}$	average daily predictive strength
IC-IR	$\overline{IC} / \text{std}(IC_t)$	risk-adjusted consistency
Newey–West t	mean IC over its HAC (25-lag) s.e.	significance under serial correlation
hit rate	fraction of days with $IC_t > 0$	sign robustness check
signal decay	$\mathbb{E}[IC(F_{t-l}, R_t)]$ vs. lag l	shelf life of the signal
factor turnover	$\mathbb{E}[\text{corr}(F_{t-l}, F_t)]$	trading cost proxy

To ensure the integrity and soundness of our results, three procedures guard every comparison in the study.

HAC significance. Because the target is a 21-day overlapping return, daily ICs are strongly serially correlated and uncorrected standard errors overstate significance several-fold. Every t -statistic in this report is Newey–West with 25 lags, wide enough to absorb the overlap established in Section 3.2.

Purged cross-validation. All in-sample method comparisons use 5-fold cross-validation on contiguous chronological day blocks, never splitting a day’s cross-section, with a 21-day purge and embargo around each test block so no training label’s forward window reaches into test days. Random row splits on overlapping labels are textbook leakage, and blocked splits with purging avoid it. Every tuned hyperparameter (λ , ρ , PLS- k , tree count) is chosen on an inner chronological validation slice of the training window, by validation rank IC, never on the test fold.

Paired tests, out of fold. A method never gets credit for its in-sample fit. Claims of the form “ A beats B ” are adjudicated by a paired Newey–West test on the out-of-fold daily IC difference series $IC_t^A - IC_t^B$: same days, same folds. This discipline pays for itself early. XGBoost’s apparent in-sample IC edge (0.047 vs. 0.031) evaporates the moment it is scored out of fold (Section 6.1).

Finally, the entire in-sample study was conducted before the blind hold-out was delivered; Section 7 describes the freeze protocol under which it was scored exactly once.

5 Methods

The candidate methods are organized from simplest to most complex, and each addition must justify itself against the simpler alternatives under the framework of Section 4.

5.1 Baselines and linear weightings

- *Benchmark.* Flip each variant by the sign of its train IC, then equal-weight average the ranks. The rank-and-average composite of Section 1.
- *PCA PC1.* The first principal component, oriented target-free to agree with the benchmark, which carries the cluster’s one train-IC sign bit (Section 3.3). The unsupervised straw man, retained precisely so its failure is measured, not assumed.
- *IC weighting.* Weights proportional to each variant’s train-window IC: weight by signal, ignore redundancy. For standardized features this coincides with the first partial-least-squares direction. In our testing it is applied both to raw features and to the cluster blocks of Section 5.2.
- *Uniqueness regression.* Weights proportional to $(\Sigma + \lambda I)^{-1}\hat{i}$, where Σ is the correlation matrix of the variants, $\hat{i}$ is the vector of their train-window ICs, I is the identity matrix, and $\lambda \geq 0$ is a ridge penalty. This is a regularized GLS direction that down-weights redundant variants, the literal implementation of “weight by uniqueness and information content.” Both a tuned- λ and a fixed- λ version are tested.
- *OLS, ridge, lasso.* Textbook cross-sectional regressions (predict, then rank), with leak-free inner-validation tuning.

- *PLS*. Partial least squares with the number of components k tuned on inner validation, asking how many supervised directions the cluster holds.
- *Boosted trees*. XGBoost and LightGBM in a deliberately low-capacity configuration (depth 3, slow learning rate, only the tree count tuned), asking whether there is structure a linear combination cannot see.

5.2 Cluster-then-weight composites

The second family tackles duplication directly: cluster the variants by average-linkage hierarchical clustering on $1 - |\text{corr}|$, merging variants whose absolute correlation is at least a threshold ρ , collapse each resulting redundancy block to its sign-aligned average, then weight the *blocks*. Three block weightings are tested: equal weights, weighting by block train ICs, and a GLS solve across blocks. The clustering step is the hard form of an equal-weight prior. Rather than penalizing weight dispersion continuously (a tuned diversification penalty, also tested, also failed), it injects the structure we actually believe: the same idea implemented twice should count once. The merge threshold is fixed at $\rho = 0.7$, at which two merged variants share roughly half their variance. Sections 6.6 and 7 show that leak-free tuning of ρ , offered repeatedly, never beats the fixed value. Figure 3 draws the four merge trees with this cut in place: across the classes, 52 variants collapse to 23 ideas.

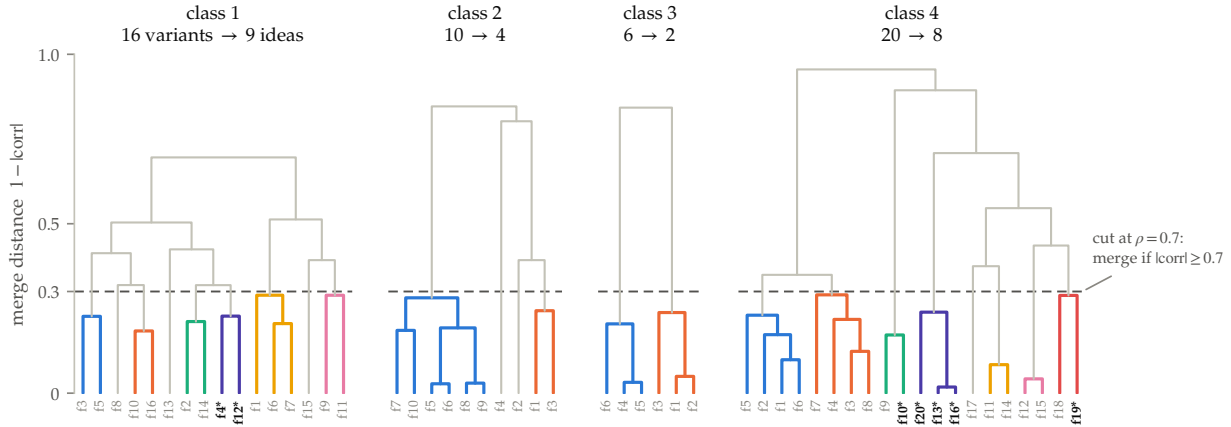


Figure 3: The recipe’s merge trees. Average-linkage dendrograms on $1 - |\text{corr}|$ for each class, computed on the sign-aligned in-sample panel; the dashed line is the fixed merge threshold $\rho = 0.7$. Colored subtrees are the resulting redundancy blocks (colors distinguish blocks within a class); gray leaves reaching the cut alone are already-unique ideas. Across the four classes, 52 variants collapse to 23 ideas. Asterisks mark sign-reflected variants (Section 3.3); the tree itself uses absolute correlation and is orientation-invariant.

5.3 A tuning-free adaptive weight, borrowed from medicine

Two weighting regimes emerge from the in-sample results below: equal block weights fit one-factor clusters, and IC-tilted weights look better where blocks truly differ. Choosing between them by validation search mostly measures luck. What we want instead is a weight that adapts by estimation, and that is a solved problem in another field.

Combining k noisy estimates of a common quantity is what meta-analysis does in clinical medicine: each study i reports an effect estimate $\hat{\theta}_i$ with standard error se_i , and the analyst must pool them. The mapping onto our problem is exact:

meta-analysis	feature composition
study i	redundancy block i (or variant i)
effect estimate $\hat{\theta}_i$	block's train-window mean daily rank IC
within-study s.e. se_i	Newey–West (25-lag) s.e. of that mean
duplicate publications of one trial	re-implemented definitions of one idea
fixed-effect pooling ($w_i \propto 1/se_i^2$)	$\approx$ IC-weighting
forcing equal w_i	the benchmark

The import is the *random-effects* model of [DerSimonian and Laird \(1986\)](#) (henceforth DL). It assumes the true per-block effects scatter with variance τ^2 (the “heterogeneity”) around a grand mean, and estimates τ^2 in closed form from the observed dispersion:

$$Q = \sum_{i=1}^k w_i (\hat{\theta}_i - \bar{\theta})^2, \quad \hat{\tau}^2 = \max\left(0, \frac{Q - (k-1)}{\sum_i w_i - \sum_i w_i^2 / \sum_i w_i}\right), \quad w_i = 1/se_i^2. \quad (2)$$

Here k is the number of blocks, $\hat{\theta}_i$ is block i 's train-window mean daily rank IC, se_i is its Newey–West standard error, w_i is the corresponding precision weight, $\bar{\theta} = \sum_i w_i \hat{\theta}_i / \sum_i w_i$ is the precision-weighted grand mean, and Q measures how much the block ICs disperse around that mean relative to their own noise. Each block IC is then shrunk toward the grand mean and the blocks are weighted by the shrunk values:

$$\tilde{\theta}_i = \bar{\theta} + \lambda_i (\hat{\theta}_i - \bar{\theta}), \quad \lambda_i = \frac{\hat{\tau}^2}{\hat{\tau}^2 + se_i^2}, \quad \text{block weight} \propto \tilde{\theta}_i, \quad (3)$$

where $\lambda_i \in [0, 1]$ is the reliability factor of the James–Stein / empirical-Bayes form: it is 0 when the estimated heterogeneity is zero and approaches 1 when a block's IC is measured much more precisely than the blocks differ.

The estimator delivers the adaptivity we wanted with zero tuned hyperparameters, and for a simple reason. The statistic Q asks one question: is the spread of the block ICs larger than their own standard errors would generate if every block carried identical information? If not ($\hat{\tau}^2 = 0$), we cannot tell the blocks apart, every λ_i is 0, and equal weights fall out automatically. If yes ($\hat{\tau}^2 > 0$), the differences are real signal, the shrinkage releases them, and the weights tilt toward the higher-IC blocks. The estimator weights by measured information content precisely to the degree the measurement itself is trustworthy. Econometrics knows the same phenomenon as the forecast-combination puzzle, and medicine got there first with a closed form.

The resulting candidate, referred to below as the *adaptive-weight* variant of the dedup composite, runs this estimator across redundancy blocks: deduplicate first, then pool, the correlated-studies discipline of the meta-analysis handbooks. A sibling variant runs it on raw variants instead, deliberately re-admitting the duplicate-publication bias so its cost can be measured.

5.4 Learned feature shapes

Every method so far consumes the variants as delivered: a stock's percentile rank enters the composite untouched. That carries a silent assumption, namely that a gap in rank means the same thing everywhere in the ranking, so that moving a stock from the 50th to the 55th percentile is exactly as informative as moving it from the 93rd to the 98th. Shape learning asks whether that assumption is true, by letting the data redraw the curve each variant is read through and seeing whether it comes out straight. The instrument is the simplest tree there is, a stump. A depth-1 stump is a one-variant, one-threshold rule (is this variant's percentile above 0.83? if so, nudge the

prediction up a step), and because every stump touches a single variant, a boosted ensemble of stumps is exactly a generalized additive model,

$$f(x) = c + \sum_{j=1}^N g_j(x_j), \quad (4)$$

where c is a constant and each g_j is a one-dimensional response curve: all the steps that reference variant j , stacked into one staircase, extractable from the fitted ensemble in closed form as a lookup table. The tree probes (Section 6.5) locate XGBoost’s one real contribution in precisely these *learned additive per-feature transforms*, and the shaped composite keeps only the curves. Each variant is passed through its learned curve as a preprocessing step, and the cluster composite of Section 5.2 is then applied to the transformed variants unchanged:

$$\tilde{x}_j = g_j(x_j), \quad F_t^{\text{shaped}} = F_t^{\text{dedup}}(\tilde{x}_1, \dots, \tilde{x}_N). \quad (5)$$

Because the shapes represent 600 boosting rounds of extra capacity, they are adopted per class only behind a mechanical *evidence gate*, defined from in-sample quantities only, with two stages and no tuned constants. Writing t_{NW} for the paired Newey–West statistic of the shaped arm’s out-of-fold daily IC against the unshaped arm’s, and ret_5 and turn_5 for the in-sample 5-day IC retention and 5-day factor turnover, a class adopts shapes only if

$$(A) \ t_{\text{NW}} > -2 \quad \text{and} \quad (B) \ \text{ret}_5^{\text{shaped}} > \text{ret}_5^{\text{linear}} \quad \text{and} \quad \text{turn}_5^{\text{shaped}} < \text{turn}_5^{\text{linear}}. \quad (6)$$

Condition (A) is a non-inferiority bar on accuracy under the purged CV, and condition (B) demands a strict improvement on both tradeability measures at once. Condition (B) encodes an asymmetry a plain accuracy comparison misses: shapes are only worth their extra capacity where the linear composite actually churns.

6 In-sample results

This section reports the in-sample study in the order it was actually run, because each step’s result determined the next. Section 6.1 establishes, on the largest class, that every sensible weighting hits the same accuracy ceiling. Section 6.2 maps all four classes and shows that the interesting differences between them are about dimensionality, not accuracy. Section 6.3 then tests the deduplication idea that this map motivates, and Section 6.4 the adaptive weighting. Section 6.5 extracts the one thing the trees add, and Section 6.6 stress-tests the resulting method before the hold-out. The summary to keep in mind throughout: accuracy differences between reasonable methods are tiny, and the real separations appear in robustness and trading behavior.

6.1 The accuracy ceiling

On class 1 (the largest, strongest cluster) every sensible linear weighting lands in a band a hair wide: the benchmark at 0.0309, IC weighting 0.0309, ridge 0.0308, tuned PLS 0.0308, the uniqueness regression 0.0306, indistinguishable inside fold noise. The two escalations built to break this tie instead certified it.

- PLS finds exactly one predictive direction. The tuner picks $k = 1$ in all five folds, and the out-of-fold component sweep peaks at one component and declines beyond it (Figure 5, first panel): PLS searched for additional supervised directions and found that none generalize.

purged 5-fold CV mean IC · shading scaled within each class,
bold = class best

benchmark	0.0309	0.0208	0.0124	0.0165
IC weighting	0.0309	0.0173	0.0108	0.0198
ridge	0.0308	0.0090	0.0141	0.0160
OLS	0.0281	0.0057	0.0148	0.0143
PLS (tuned)	0.0308	0.0133	0.0146	0.0163
uniqueness regression	0.0306	0.0132	0.0114	0.0187
PCA PC1	0.0305	0.0183	0.0125	0.0033
boosted trees	0.0315	0.0095	0.0048	0.0121
	class 1	class 2	class 3	class 4

Figure 4: Purged 5-fold CV mean IC for every method on every class. Shading is scaled within each class column (darker = higher IC on that class) and the bold entry marks the class best. The benchmark is the robust anchor everywhere, supervision helps only where the cluster is diffuse (class 4), and trees collapse on the three smaller or tighter classes.

The one-component PLS weight vector is the IC vector, so PLS and IC weighting coincide here by construction.

- Trees find no nonlinear structure worth the name. Leak-free tuned XGBoost lands at 0.0315, +0.0006 over IC weighting, a tenth of its own across-fold standard deviation, and its inner tuner prefers minimal capacity (three of five folds want ≤ 50 boosting rounds). Its in-sample IC of 0.047 collapses to 0.031 the moment it is scored out of fold.

The uniqueness regression, the literal implementation of the project’s objective, fails on this cluster for a reason the regularization path makes visible: out-of-fold IC improves as λ grows, that is, as the method stops trusting the off-diagonal of the estimated correlation matrix. Class 1 is nearly one-dimensional, so no variant is meaningfully more unique than another, and the matrix-inverse correction amplifies estimation noise in exactly the small-eigenvalue directions where noise lives. The bias–variance ledger favors the simpler method. This is a negative result about this cluster, not the method. The real test is class 4, flagged in the exploratory analysis as having several sub-themes.

6.2 Variance dimension versus predictive dimension

Having pinned the class-1 ceiling, I ran the full suite on all four classes (Figure 4). The pattern generalizes cleanly. Each class has its own ceiling (0.0315 / 0.0208 / 0.0148 / 0.0198 for classes 1–4), the benchmark is at or within noise of the best method everywhere (it literally wins class 2), and supervised weighting earns a visible point edge exactly where the cluster is diffuse (class 4: +0.0033 via IC weighting). Meanwhile the tree result reverses off class 1: XGBoost loses between a quarter and two-thirds of the benchmark’s IC on classes 2–4 (54%, 61%, and 27% on classes 2, 3, and 4). Its class-1 parity was the best case (large panel, strong signal), and LightGBM reproduces both the parity and the collapse, making the verdict generic to boosted trees.

The sharper instrument is Figure 5: an out-of-fold PLS- k sweep per class, measuring how many predictive directions each cluster has, next to its eigen-spectrum measuring variance directions. The two disagree in both directions, and the disagreements are invisible to any unsupervised analysis.

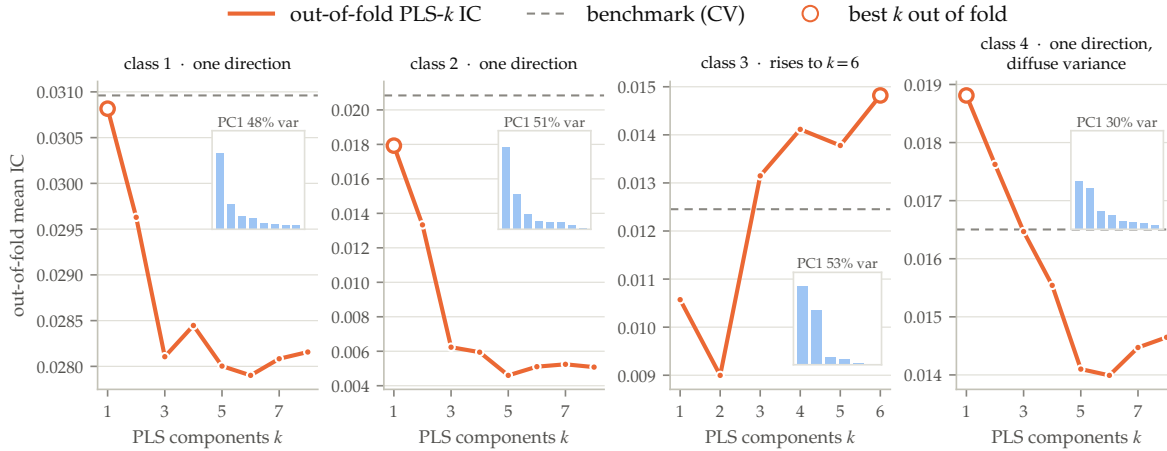


Figure 5: Predictive dimension per class: out-of-fold PLS- k IC (line, best k circled; the benchmark’s CV IC dashed for reference) and the feature eigen-spectrum (inset bars). Classes 1 and 2: one direction. Class 4: one predictive direction hiding in a diffuse variance structure. Class 3: predictive content spread across many low-variance directions.

- Class 4 looks multi-factor but predicts one-dimensionally. Variance is spread across many directions (PC1 only 30%), yet everything predictive lies along the first supervised direction. IC weighting’s class-4 edge is not about combining several predictive factors. It is about finding the one predictive direction when PC1 no longer points at it.
- Class 3 is the reverse: two variance blocks, many predictive directions. Out-of-fold IC keeps rising to $k = 6$. Later, low-variance directions carry real incremental information, which is why regression-style fits (OLS, small- λ GLS) top this class.

These profiles are the study’s central diagnostic reading, and they dictated what to build next. Class 4’s profile says the valuable operation there is not estimating clever weights but identifying which variants repeat one idea, so that the single predictive direction stops being drowned by the most-duplicated definition: exactly deduplication, no regression required. Class 3’s profile says the opposite temptation (chase the extra directions with a fitted k or a GLS solve) is real but rests on one free parameter estimated from a weak class, and Section 7.3 shows what honoring that temptation costs when the fit is frozen. Classes 1 and 2 say: any sensible weighting works, so spend the complexity budget elsewhere or not at all. One route satisfies all four readings at once, cluster the redundancy away and weight the surviving ideas equally, and that is the route the next section takes.

6.3 Deduplication

The cluster-then-weight family, cross-validated on all classes with paired out-of-fold Newey–West tests against both anchors (the benchmark and IC weighting), delivers the study’s cleanest positive and negative findings.

1. Dedup with equal weights, cluster then count each idea once, is the only method that never loses to the benchmark anywhere, and the only one that ever significantly beats it. On class 4 it gains +0.0028 daily IC over the benchmark with Newey–West $t = 2.26$, using no target information beyond sign bits. Supervised rivals post slightly larger point gains there (+0.0033/+0.0036) with lower significance ($t = 1.82/1.63$): estimated weights add point-estimate gains and variance in roughly equal measure. The central premise is confirmed out

of fold. The benchmark’s flaw is duplication, not equal weighting, and fixing only duplication captures the achievable gain at zero supervision cost.

2. Supervision is a measurable tax wherever the cluster is one clean factor. On class 2 every target-using weighting (IC weighting, the uniqueness regression, and its tuned version) is significantly worse than the benchmark, with deficits of 35 to 117 basis points of daily IC (1 bp = 0.0001) at t between -2.4 and -2.9 . One predictive direction plus noisy daily ICs means estimated weights are noise around the equal-weight optimum: the forecast-combination puzzle, reproduced inside a single signal.
3. Adaptivity by search fails, and so does the soft prior. Letting inner validation pick the method per class is never terrible and never optimal. Model selection is itself an estimation problem: candidates differ by 0.002–0.004 against inner-slice noise an order larger. A tuned shrink-toward-equal-weights penalty never wins either. The hard version of the same prior (clustering) beats the soft version (a penalty), because it injects structure rather than another free parameter.
4. Nothing is hiding in the estimation window. Rolling and daily re-estimation (including literal daily PCA from each day’s ~ 500 -name cross-section) only lose. The redundancy structure is time-stable, so trailing windows add estimation noise and remove nothing. Pool everything.

Because the one significant win sits only just past the 2σ line, I also checked its sensitivity to the concordance measure within the rank family: re-scoring the identical out-of-fold signals under Kendall’s τ , which no single name can dominate, the class-4 win holds at $+0.0019$ with Newey–West $t = 2.29$, marginally stronger than under Spearman, and every t in the eight-row check moves by at most 0.04. Both measures are rank-based by design (Section 4). What the check establishes is that the win is broad concordance across the cross-section rather than a handful of influential names, not that the result is independent of the rank-based scope itself.

6.4 The heterogeneity estimator as a detector

Section 5.3’s estimator was imported to make the equal-versus-IC-weights decision by estimation rather than search, and it delivered as a *detector*. Estimated per class and per fold from training data alone, the heterogeneity parameter came back $\hat{\tau}^2 = 0$ (to display precision) in all five folds on classes 1–3, prescribing full pooling into equal weights, and $\hat{\tau}^2 > 0$ in all five folds on class 4, prescribing that the IC differences be kept. That is precisely the supervision map Sections 6.2–6.3 needed a battery of paired tests to establish, recovered by one closed-form formula with nothing to tune. The estimator’s choices also paid directly. On class 2 the DL-weighted composite dodges the supervision tax automatically and beats IC weighting significantly ($+0.0036$, $t = 2.52$), and the adaptive-weight variant of the dedup composite matched the per-class best method everywhere with zero tuned weighting decisions.

Two caveats were recorded at the time, and both mattered later. First, the adaptive variant’s edges over the plain equal-weight form were never individually significant. Second, detection and monetization are different claims: $\hat{\tau}^2 > 0$ says class-4 blocks truly differ in information content, but it does not guarantee that weights fitted to those differences transfer forward. The blind hold-out confirmed exactly this gap (Section 7), and the lesson, keep the detector and drop the tilt, became the final method’s form. We also tested a natural two-level extension, shrinking each variant’s IC toward its block mean rather than collapsing the block outright: offered any pooling depth, the estimator chose full pooling in 19 of 20 class-folds. Within a redundancy block, member

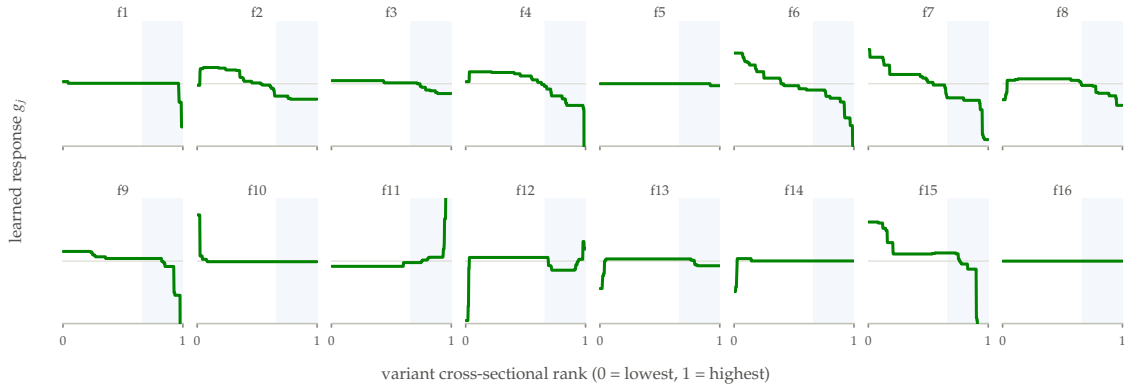


Figure 6: The learned per-feature transforms, extracted exactly from the depth-1 ensemble (each panel on its own vertical scale; the shaded band is the top third of ranks). The curves are flat through the middle of each variant's rank range and respond sharply in the extreme tail: 75% of total response movement lives in the top third, 6% in the middle. The model reads class 1 as a tail-depth detector, and the production model is 16 lookup tables, no tree library required.

ICs are statistically indistinguishable. The idea was not wrong, it was already embedded in the method.

6.5 What the trees contribute

XGBoost's one out-of-fold-verified edge on class 1 was not accuracy but trading behavior: 5-day IC retention of 0.72 against roughly 0.60 for the linear composites, a half-life of 11.1 days against 6.8, and the lowest 5-day factor turnover of any method. Depth-1 stumps, which cannot represent interactions, reproduce the full tree's entire profile, and no construction on the linear composite does. The edge is therefore learned additive per-feature response curves, extractable from the stump ensemble in closed form as 16 lookup tables (Figure 6). The curves are flat through mid-ranks, where daily rank jitter lives, and respond only in the extreme tail, a slowly-changing state. That is the turnover mechanism, and a two-compartment decay fit prices it: the share of the signal decaying on the slow timescale rises from 0.29 for the linear composite to 0.54 for the shaped one. In effect the data redraw the ruler each variant is measured with, and the ruler it drew ignores mid-pack shuffling and counts only depth into the tail: the composite says the same thing it always said, it just stops changing its mind over noise.

Grafting the shapes onto the dedup composite produced the study's most useful single object. On class 1 the shaped composite reaches a half-life of 13.5 days, retention of 0.73, and turnover of 0.36 in-sample, out-trading the tree that taught it the shapes, at the linear ceiling's accuracy. On classes 2–4, applying shapes unconditionally is somewhere between pointless and disastrous (class 2: -0.0180 , $t = -2.97$). Six hundred stumps on 1,500 days repeats the tree collapse of Section 6.2, which is exactly why shapes only enter behind the evidence gate of Section 5.4. The gate adopts class 1 alone.

6.6 Stress tests

Before the hold-out, I attacked the dedup composite from every direction I was able to think of, as well as those raised in sponsor meetings. Two examples:

- Partition construction. Five linkages, an IC-profile similarity, a top-down divisive tree, and a random placebo, all at matched block count: the partition matters only on class 4, where

the anchor tree wins and the placebo collapses (-0.0025). It is irrelevant by construction wherever $\hat{\tau}^2 = 0$ forces equal block weights.

- Tuned merge threshold. Per-fold inner-validation tuning of ρ over a wide grid confirms 0.7 (class 4: five folds of five), picks a stable neighbor that buys nothing (class 1: 0.8 in five of five, $+0.0006$, not significant), or whipsaws unstably (classes 2–3) and costs money. The sensitivity table’s best cell (class 2 at $\rho = 0.8$, ten basis points above the selected value) is not ex-ante selectable: leak-free tuning delivered 0.0208, not 0.0226. The gap between the best cell in a sensitivity table and what tuning actually delivers is this study’s recurring look-ahead lesson.

The test that ended up anchoring my confidence came from perturbing the feature set itself. The premise of the whole project is that the exact list of variants a class contains is arbitrary at the margin: a team could easily have delivered 12 of these 16 variants rather than all 16, so a composition method must not depend on the exact list it happened to receive. I therefore re-ran the full purged CV on 80 randomly reduced feature sets, 20 per class, each formed by dropping roughly 25% of that class’s variants at random, and scored every method on every subset (Figure 7). The result is the study’s clearest separation. The dedup composite ranks top-2 in mean rank on classes 1–3 and mid-pack on class 4 (behind the two IC-tilted arms), and its worst case is the shallowest of anything tried, jointly with its adaptive-weight sibling (Figure 8, panel B). Every rival craters somewhere: the benchmark is last on 17 of 20 class-4 subsets with the lowest floor there, since dropping the right variants leaves its duplication bias nothing to hide behind; tuned uniqueness collapses on class 2 (its worst subsets sit at roughly 40% of the benchmark’s IC), and the GLS edge on class 3 inverts under perturbation. Robustness of this kind is invisible in any single full-set comparison, and it is the property a desk actually needs: the composite never needs to know which class, or which subset of a class, it is running on. That property, not any mean edge, is why the multiplicity verdict of Section 8 (“the floor, not the mean”) matches the method’s natural defense.

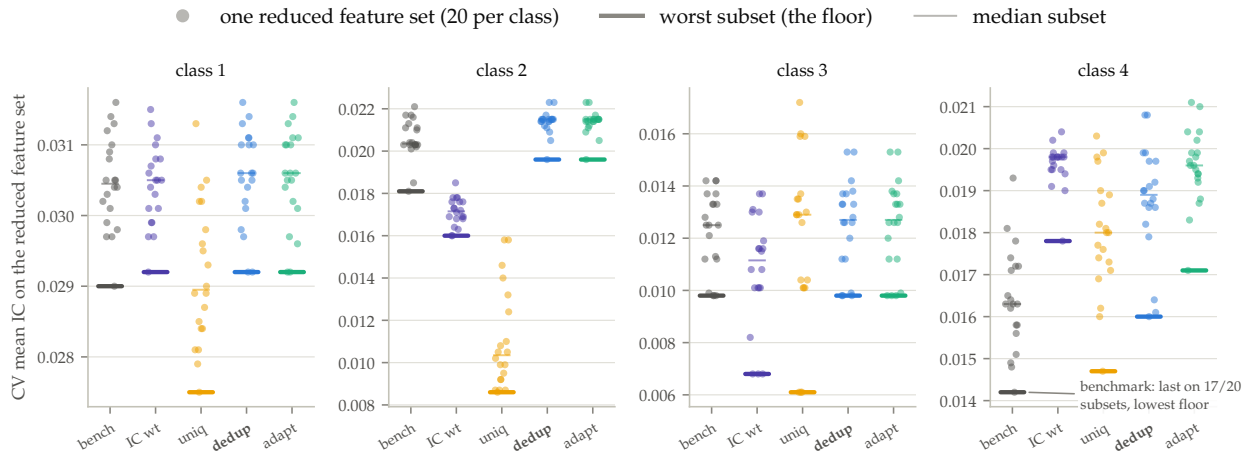


Figure 7: The feature-subset stress test: 80 randomly reduced feature sets (20 per class, $\sim 25\%$ of variants removed, full purged CV re-run per subset). Each dot is one reduced subset’s CV mean IC for one method; colors identify methods, consistent with the other figures. The heavy tick is each method’s worst subset, its floor, and the thin tick the median. The dedup composite (“dedup”) holds the highest or joint-highest floor on classes 1–3 and the joint-shallowest worst-case shortfall overall. The benchmark is last on 17 of 20 class-4 subsets with the lowest floor there, and tuned uniqueness craters on classes 2 and 3.

7 The blind hold-out

7.1 Protocol

The hold-out arrived after the in-sample study was complete: the immediately following 2,195 calendar-indexed trading days for every class ($\sim 2,180$ scoreable days, ~ 430 names per day). The protocol is strict freeze. Every parameter (feature signs, IC weights, tuned λ and ρ , the clustering, the adaptive block weights, the XGBoost model, the stump shapes and the gate decision, even standardization constants) was estimated once on in-sample data and applied unchanged. Nothing is fit, tuned, or selected on the hold-out. One further lock was applied: after the final in-sample review confirmed that tuning ρ yielded negative results and that no universal fixed ρ significantly beats 0.7, we fixed it at 0.7. The textbook regressions (OLS, ridge, lasso) were retired in-sample due to their unpromising results and not frozen for hold-out scoring. The tuned uniqueness arm stands in place for the regression family below.

7.2 Results

Figure 9 is the scorecard and Figure 8 the study in one figure, and the full numeric table is Appendix A. Three headlines order the results. First, the signals are real out of sample: classes 2–4 are significantly positive under strict freeze, with Newey–West t between 2.4 and 3.9 for the benchmark and dedup composites. Second, the accuracy claim replicates exactly as stated, which is to say at zero: the dedup composite minus the benchmark is approximately zero everywhere (max $|t| = 1.1$ for the linear form, 1.3 including the shaped class-1 arm), while the composite matches or exceeds the benchmark on every class. Third, everything the study had flagged as fragile is precisely what failed. Three results deserve their own paragraphs.

Class 1 is the alpha-decay lesson. Its linear composites lost two-thirds of their IC ($0.031 \rightarrow 0.0119$, $t \approx 1.9$), yet the drift diagnostic shows the redundancy structure barely moved. The correlation matrix’s Frobenius drift is 0.22, in line with the other classes, not one of 16 variants flips IC sign, and all shrink uniformly 50–70%. The correlation matrix is stable; the signal level decayed. Figure 10 shows how: the rolling one-year IC sits near, mostly slightly below, the in-sample level for most of the eight-year window and breaks downward in the final year, while classes 2–4 cycle around positive means. Nothing re-weightable was lost. A rolling re-estimation arm (quarterly refits on an expanding window) recovers exactly nothing on class 1, because there is nothing stale to refresh. The operational moral for any frozen composite is to monitor the level, not just the structure, because alpha decay cannot be recovered by re-estimation.

The shaped composite is the class-1 method that survives. Every linear composite sits at $t \approx 1.9$ on class 1. The shaped composite is the only linear-family composite still significant there, and against the one other survivor, the tree that taught it the shapes, it matches on accuracy and wins on every tradeability axis:

class 1, hold-out	mean IC	NW t	5-day retention	half-life	5-day turnover
shaped composite	0.0154	2.45	0.62	15.0d	0.39
boosted trees	0.0149	2.47	0.54	7.0d	0.57
linear dedup / benchmark	0.0119	1.9	0.37	2.6d	~ 0.6

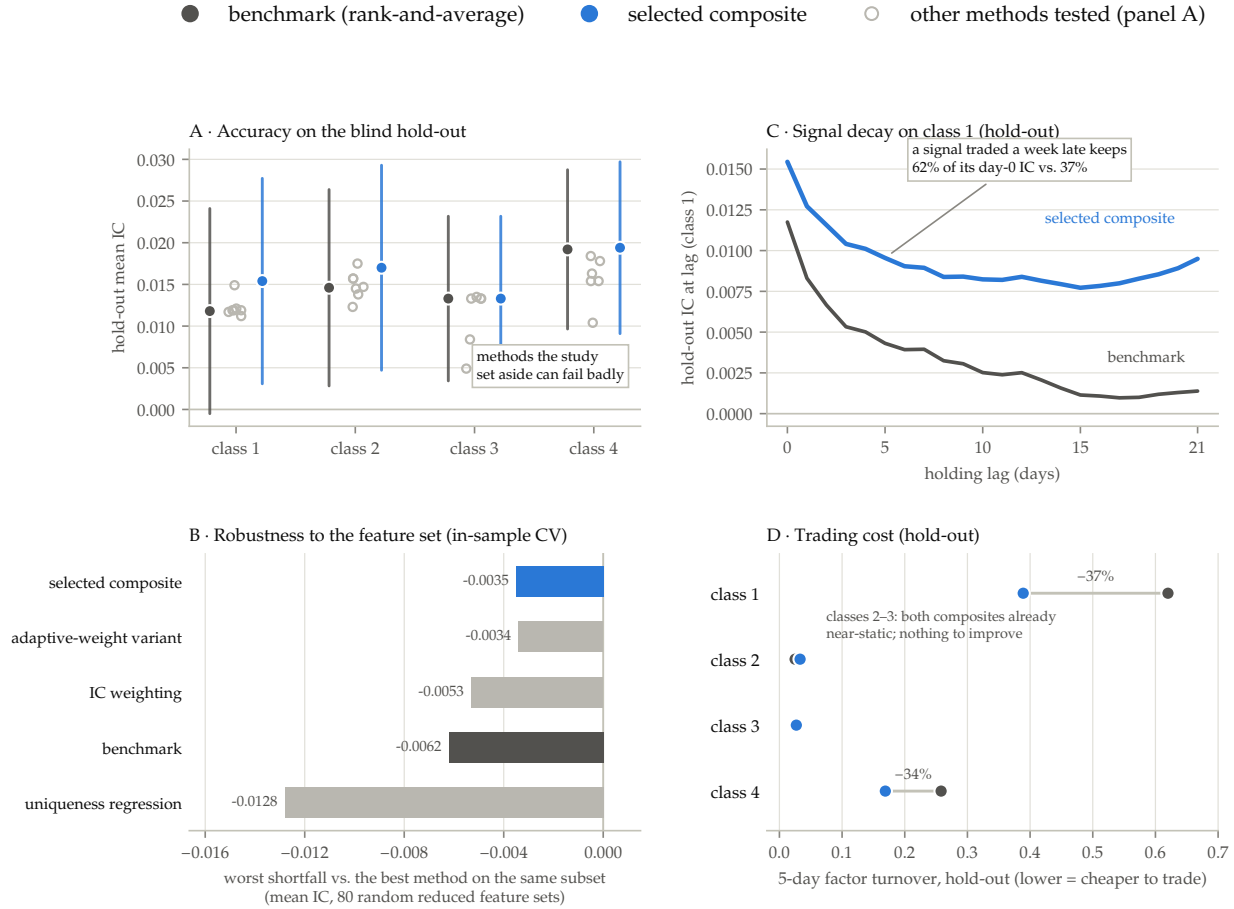


Figure 8: The study in one figure. Throughout, gray marks the benchmark, blue the selected composite (on class 1 this includes the learned feature shapes), and small open gray circles the other frozen methods. (A) Hold-out mean IC per class with 95% Newey–West confidence whiskers: the selected composite matches or exceeds the benchmark everywhere, and the low gray circles on classes 3–4 show how badly the discarded methods (PCA, tuned PLS, boosted trees) can fail. (B) The one in-sample panel: across 80 randomly reduced feature sets (Section 6.6), each bar is a method’s worst shortfall against the best method on the same subset, and the selected composite’s worst case is the shallowest. This test is pre-freeze by construction, since re-running method selection on the hold-out would contaminate it. (C) Hold-out IC by holding lag on class 1, the class where the composites differ most: the selected composite starts higher and decays far more slowly. (D) Five-day factor turnover per class: lower for the selected composite where the two composites differ (classes 1 and 4), unchanged where they coincide.

The shaped-minus-linear IC gap ($+0.0035$, $t = 1.3$) is not itself significant. The case for shapes is IC parity plus the trading profile, priced jointly in Section 8. The pre-specified sign-aligned refinement of the shaped arm (flip each block member by the sign of its train IC before averaging) ties the original everywhere (0.0155 vs. 0.0154): the tidy-up was safe, not material.

The adaptive tilt is the casualty. The adaptive-weight variant trails plain equal weights by -0.0031 ($t = -3.4$) on class 4, the one class where the weights differ. Quarterly re-estimation lifts the tilt only to 0.0182, still below frozen equal weights at 0.0194: tilt plus maintenance loses to equal weights frozen forever. The detector stays and the tilt goes. From here on the selected

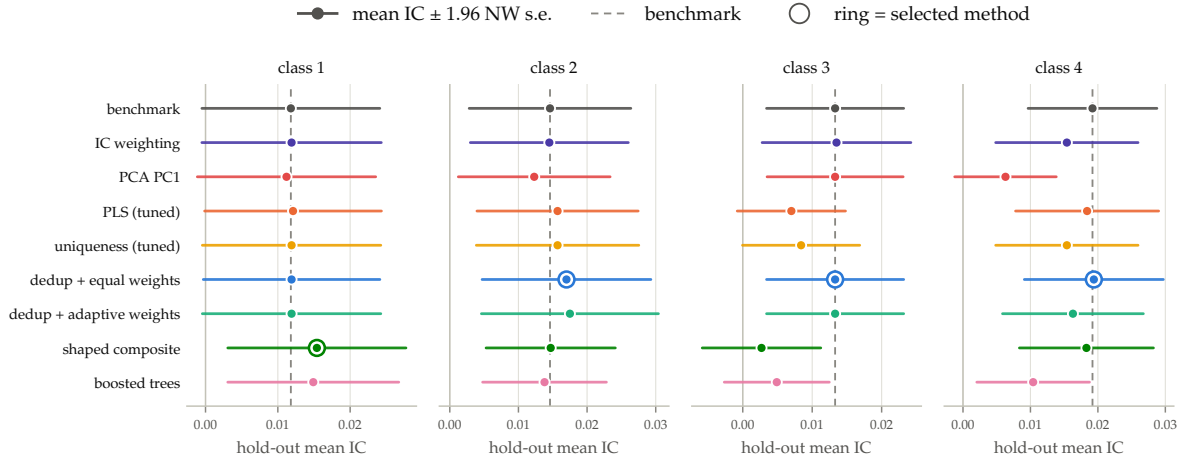


Figure 9: The frozen methods on the blind hold-out ($\sim 2,180$ unseen days per class): mean daily rank IC with ± 1.96 Newey–West standard errors. The dashed line is the benchmark’s level, and the ring marks the method selected for each class. The selected composite matches or beats the benchmark on every class. Classes 3 and 4 realized above their in-sample CV level, a period effect at the year scale (Figure 10). The two instructive failures are visible at a glance: PCA collapses on the multi-factor class 4, and tuned PLS and the shaped arm both break on class 3.

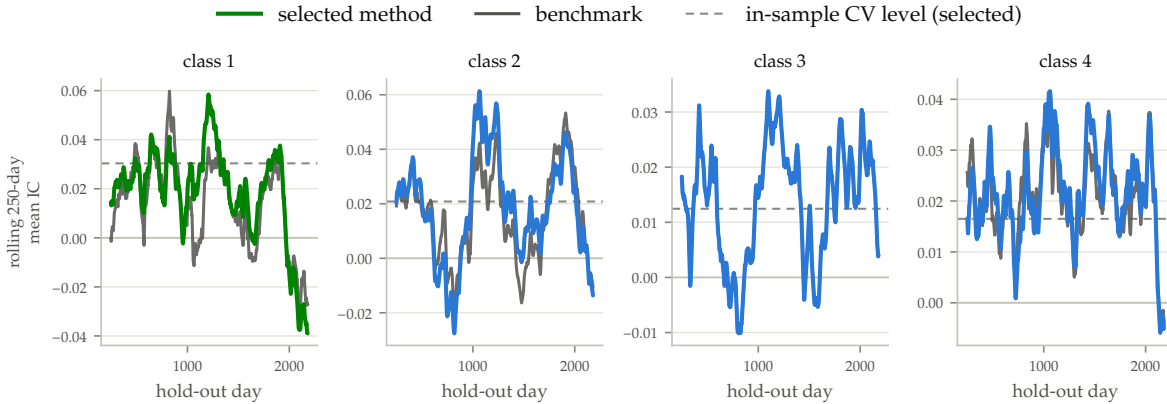


Figure 10: The hold-out IC through time: rolling 250-day mean IC of the selected method (green) against the benchmark (gray), per class, with the selected method’s in-sample CV level dashed for reference. Class 1’s loss concentrates in the final year of the window, while classes 2–4 cycle with year-scale swings around positive means.

composite means the equal-weights form, with the adaptive variant demoted to a candidate the blind test retired.

7.3 PCA and PLS out of sample

Because the in-sample study retired PCA and tuned PLS on in-sample evidence alone, both were frozen post hoc under the identical protocol and scored: a test of the retirement decisions themselves on data neither method ever saw (Figure 11).

For PCA, retirement is confirmed. Where a cluster is near-one-factor, PC1 essentially is the average and adds nothing (class 3: 0.0133 vs. 0.0133, the curves coincide). On the one multi-factor class it keeps a third of the benchmark’s IC (0.0063 vs. 0.0192) while trading more and decaying

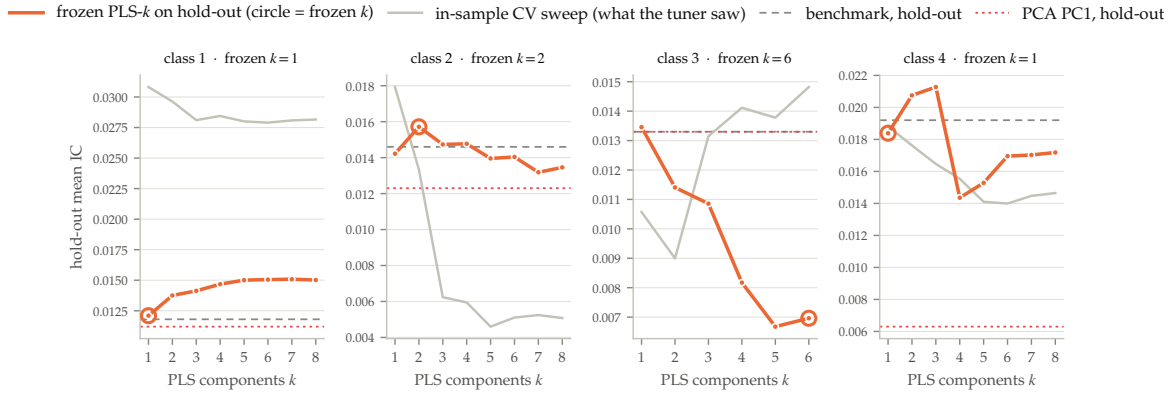


Figure 11: The frozen PLS- k sweep on the hold-out, per class. The orange line is the frozen PLS composite’s hold-out IC as a function of the number of components k , the circle marks the k chosen on in-sample inner validation and then frozen, the gray line is the in-sample CV sweep the tuner saw, and the dashed lines are the frozen benchmark and PCA PC1 hold-out levels for reference. On class 3 the in-sample sweep rises to $k = 6$ while the hold-out falls with k instead: the inner-validation choice halves the class’s IC. Because k is estimated in sample, estimation noise means the out-of-sample optimum can sit at a starkly different k ; the intended benefit is lost in the act of estimating.

faster. As a four-class package it posts the study’s only $|t| > 3$ pooled result, a deficit (Section 8). Out of sample as in sample, variance is the wrong objective.

PLS is the sharpest exhibit of the study’s recurring pattern. Where the tuner froze $k = 1$ (classes 1 and 4), one-component PLS approximately equals IC weighting, as the algebra requires, and on class 2 its frozen $k = 2$ is the one tuning win (+0.0011, not significant). But on class 3 the tuner froze $k = 6$, a defensible choice given that the in-sample out-of-fold sweep really does rise with k , and the hold-out falls with k instead: 0.0070 against 0.0135 at $k = 1$. One leak-free inner-validation decision on one free parameter cost the class half its IC. Supervision that wins the inner split does not survive the freeze. This is the same lesson as the tuned ρ and the monetized τ^2 : when estimation is introduced, much of the intended benefit is lost to noise.

8 Statistical checks

8.1 Multiplicity correction

The study ran 56 paired “does X beat the benchmark?” comparisons, all on the purged-CV out-of-fold record (12 methods $\times$ 4 classes, plus the retired PCA and PLS; hold-out differences are reported separately in Section 7). Seven clear $|t| > 2$ nominally: six negative and one positive (the dedup composite’s class-4 win, $t = 2.26$). Following [Harvey et al. \(2016\)](#), I correct the whole family (Figure 12). Zero tests survive Holm at 5%, Benjamini–Hochberg at 5%, or the dependence-aware Westfall–Young max- T (block bootstrap, whose 95% bar sits at $|t| \approx 3.75$; the win’s adjusted $p \approx 0.66$). At the looser BH 10% false-discovery rate, exactly two results survive, both negative: PCA on class 4 (−0.0132, $t = -3.11$) and tuned uniqueness on class 2 (−0.0117, $t = -2.93$). The only comparisons that retain any post-correction standing in the whole study are negative results.

This re-frames the study. No positive paired comparison here, including our own headline win, should be quoted as significant on its own. The defensible claim shifts to where the stress tests had already placed it, the floor rather than the mean, which multiplicity correction does not touch. The negative census, meanwhile, has teeth: do not chase variance on a multi-factor cluster, and do not supervise a one-factor cluster, is the closest thing to a certified finding this analysis

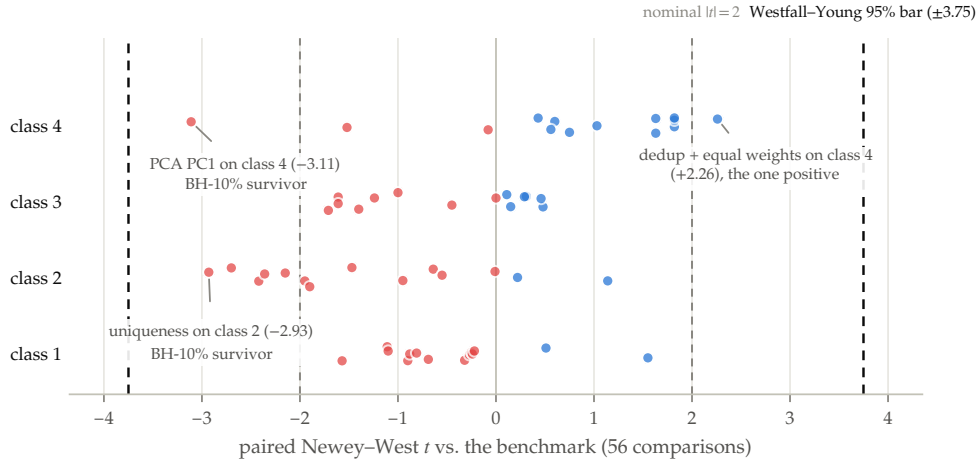


Figure 12: Every paired method-versus-benchmark t -statistic in the study (the in-sample purged-CV family), one dot per method and class, blue for positive and red for negative. The inner dashed lines mark nominal $|t| = 2$, which flags seven comparisons; the outer dashed lines mark the dependence-aware Westfall–Young 95% max- T bar (± 3.75), which flags none. The two survivors at the looser BH-10% rate are both negative results.

produced. The test has power, since it flags the known pathologies at BH 10%, so the null results elsewhere reflect an absence of effect rather than an absence of power.

8.2 Robustness of the heterogeneity detection

DerSimonian–Laird is known to be noisy at small k . Re-estimating the method’s one data-driven decision with the Paule–Mandel and REML estimators reproduces the τ^2 map, zero on classes 1–3 and positive on class 4, at the frozen full-sample level, and all three estimators find $\hat{\tau}^2 > 0$ in every one of class 4’s five CV folds. The detection is estimator-robust. But frozen on the class-4 hold-out, the three tilts all land between 0.0163 and 0.0166 while equal weights score 0.0194: whatever τ^2 correctly measured about in-sample heterogeneity, the weights it prescribes did not carry forward. The Hartung–Knapp intervals explain the failure in one number. With $k = 2$ blocks (class 3) the 95% interval on the pooled block IC is $[-0.016, +0.036]$. At these block counts the estimator can detect heterogeneity but cannot support differential weights. Detection is robust, monetization fails, and the method keeps τ^2 as a diagnostic flag with equal weights as the action.

8.3 Shape stability across feature subsets

The class-1 shape edge was learned on the 16 variants the class happened to contain, so I re-learned it on each of the twenty reduced feature subsets of Section 6.6 (per-fold stump refits in CV, then freeze-forward scoring on the hold-out). In sample, the shaped composite’s CV cost is a rounding error: mean -0.0002 , with zero significant losses in 20 subsets. Out of sample, the frozen shaped arm beats the frozen linear arm on IC on 18 of 20 subsets (mean $+0.0021$), trades less on 17 of 20, retains 0.61 of its IC at lag 5 against 0.39, and wins the weekly criterion below on 20 of 20. The gate, replayed per subset, adopts on 19 of 20. Its one refusal cost $+0.0003$ on the single subset where the shaped arm did churn more out of sample, so the screen caught a real defect cheaply. An edge that survives twenty random re-drawings of its inputs, on unseen data, in level and profile, is a property of the class, not of the exact feature list.

8.4 A weekly rebalancing criterion

The metrics so far price accuracy and trading behavior separately, but a real book earns them jointly, so the last check prices them together. Ultramarin’s holding horizon is mid-to-low frequency, in the neighborhood of days to weeks, so we take weekly rebalancing as the measurement criterion. Between rebalances such a book trades on a signal that is between zero and five days stale, so the IC it earns is approximated by the average of the lag-0 and lag-5 daily ICs. That average is the criterion used here: it captures level and decay in one number and, as a daily series, supports the same paired Newey–West testing as everything else. Against the benchmark, the only material per-class term is shapes on class 1: $+0.0047$ per day ($t = 1.6$), a 59% relative improvement (0.0127 vs. 0.0080), earned because the benchmark keeps 37% of its IC at lag 5 while the shaped arm keeps 65%. Everything else is noise-level. Pooling day-by-day across classes gives Table 2.

Table 2: The weekly level-plus-decay criterion, pooled across classes, paired against the benchmark on the hold-out. Each row is one four-class package: which composite serves each class. The retired references calibrate the scale: the criterion can flag a bad package at $t = -3$, so the selected package’s $+1.72$ is competing against a real zero, not against a test with no power.

four-class package	weekly-criterion edge vs. benchmark (per day)	NW t
shaped c1 + dedup eq. c2–4 (selected)	$+0.0019$	$+1.72$
shaped c1 + dedup adaptive c2–4	$+0.0013$	$+1.02$
dedup equal weights everywhere	$+0.0007$	$+1.05$
dedup adaptive weights everywhere	$+0.0001$	$+0.09$
<i>PCA PC1 everywhere (retired reference)</i>	-0.0039	-3.07
<i>PLS tuned everywhere (retired reference)</i>	-0.0013	-1.20

Two labels belong on this number. The t of 1.72 is suggestive, not certified, and the package combination is post hoc, although its components were pre-specified. Turnover sits on top of the criterion: the winning package trades at roughly 60% of the benchmark’s turnover on class 1 and roughly 35% lower on class 4. What makes the number credible is its consistency. The same ordering holds on every class where anything differs, on 20 of 20 class-1 feature subsets, and under every aggregation tried. Figure 13 is the graphical form: the selected package’s hold-out IC is at or above the benchmark’s at every holding lag on every class, and on the two classes where the method changed anything (1 and 4) the edge grows with the holding period.



Figure 13: Selected composite minus benchmark, hold-out IC at every holding lag, per class, on one vertical scale throughout; this is the per-class expansion of the weekly criterion. The shaded difference is at or above zero everywhere: no holding horizon, on any class, favored the benchmark over the hold-out period. On class 3 the selected composite’s blocks coincide with the benchmark, so the difference is identically zero.

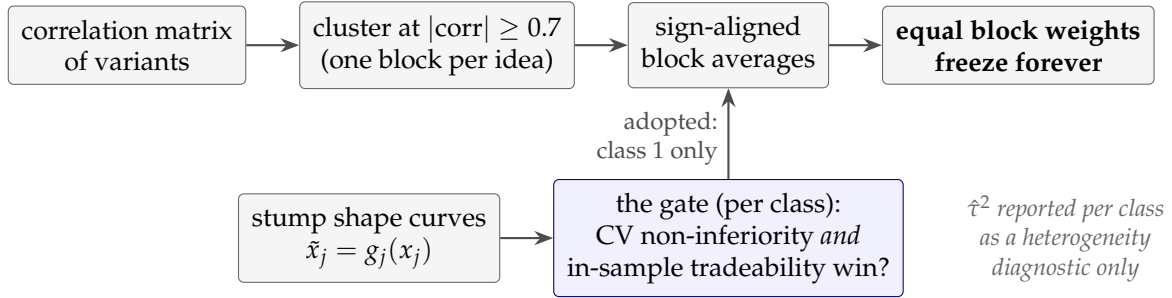


Figure 14: The selected composite. Variants are clustered into idea-level blocks, averaged within blocks, and the blocks weighted equally, with all parameters frozen. The gate is the one data-driven decision point: learned feature shapes enter only where they pass the two-stage in-sample bar of Section 5.4, which happened on class 1 alone. There are no tuned hyperparameters and nothing on a re-estimation schedule.

9 The selected composite methodology

The final specification (Figure 14), with the evidence for each choice:

1. Group the duplicates. Cluster variants on $1 - |\text{corr}|$ (average linkage); variants more than 70% correlated in absolute value merge into one block, so the same idea implemented several times gets one vote. The threshold is fixed forever: tuning was offered a wide grid twice and could not beat it, and the value has a structure-free meaning (shared variance $\approx$ half).
2. Average within blocks. Flip any variant whose train IC is negative, then simply average each block's members. A handful of idea-level signals replaces 6–20 redundant variants.
3. Weight the ideas equally and freeze. The random-effects tilt detected real heterogeneity where blocks truly differed, confirmed by three estimators, but acting on it lost to frozen equal weights on the blind hold-out, and quarterly re-estimation only partially repaired that self-inflicted loss. Equal block weights match or beat the tilt everywhere, frozen, with no refit infrastructure. $\hat{\tau}^2$ stays in the pipeline as a diagnostic, a flag that a class is multi-factor and deserves attention, not as weights.
4. Shapes, only where the evidence clears the bar. Variants may first pass through their learned response curves (sign-aligned form), but only where the gate says so: empirically class 1 alone. On that class the shaped composite is the best method on the hold-out at roughly 60% of the linear form's turnover, and the only composite in the linear family still significant there.

Table 3: The selected composite per class, with its blind hold-out scorecard. Classes 2–4 are the dedup-and-equal-weight rows throughout (retention and turnover recomputed from the stored hold-out curves). Class 1 shows the shaped blind-test record; the selected sign-aligned form is statistically identical (IC 0.0155, $t = 2.44$, retention 0.65, turnover 0.37).

class	composite	mean IC	NW t	5-day ret.	5-day turn.	hit rate
1	shaped $\rightarrow$ dedup $\rightarrow$ equal blocks	0.0154	2.45	0.62	0.39	0.57
2	raw $\rightarrow$ dedup $\rightarrow$ equal blocks	0.0170	2.71	0.94	0.03	0.60
3	raw $\rightarrow$ dedup $\rightarrow$ equal blocks	0.0133	2.64	0.97	0.03	0.60
4	raw $\rightarrow$ dedup $\rightarrow$ equal blocks	0.0194	3.69	0.89	0.17	0.62

In one sentence: *count each idea once, weight ideas equally, learn feature shapes only where the evidence is strong, and freeze everything*. Its virtue is not extra IC. After multiplicity accounting, nothing beats the benchmark on IC. Its virtue is that on the level-plus-decay criterion a real book earns, it is consistently ahead (+0.0019 per day pooled, $t = 1.7$, suggestive rather than certified, at 35–40% lower turnover where turnover matters), it has the joint-shallowest worst case of anything tried, it is fully mechanical, and every moving part carries the test that justifies it.

Equally important is what the composite deliberately leaves out, because each exclusion carries a documented failure: IC weighting (significantly hurts one-factor clusters), GLS and uniqueness solves (the class-3 edge inverted out of sample), the adaptive tilt (-0.0031 frozen), scheduled re-estimation (repairs only the tilt’s own damage), boosted trees off class 1 (27–61% IC collapse), tuned merge thresholds and search-based method selection (whipsaw), and PCA (one-third of the benchmark’s IC on the class where it matters). A desk that adopted the “obvious” improvements without this discipline would end up strictly worse than one deploying plain rank-and-average.

10 Discussion

Four anonymized classes cannot certify a universal rule; that is a sample of four. What the study can hand Ultramarine is a set of cheap, mechanical diagnostics with a measured track record, a map from diagnostic readings to actions, and a list of ideas that need exactly the resources we did not have. This section is addressed to readers who know what classes 1–4 actually are.

10.1 A playbook for a new feature class

Everything below is computable from a new class’s training panel in minutes, before any composite is deployed.

1. PC1 variance share and the redundancy dendrogram (Section 3). Above ~ 0.45 with clean blocks: equal weighting will saturate the class, and the composite without shapes is likely already optimal. A diffuse spectrum (class-4-like, PC1 ~ 0.3): the partition now matters, so inspect the dendrogram, and expect the plain benchmark to be the casualty (last on 17 of 20 class-4 feature subsets, with the lowest floor).
2. The out-of-fold PLS- k sweep (Figure 5). One peak at $k = 1$: any sensible weighting works, so spend effort elsewhere. Rising past $k \geq 3$ (class-3-like): the class carries real multi-directional signal that block averaging partially destroys, but our evidence says do not chase it with a frozen tuned k or a GLS fit (both inverted out of sample). Treat it as a research flag, not a production decision, until several such classes accumulate.
3. $\hat{\tau}^2$ across blocks, under all three estimators (Section 5.3). $\hat{\tau}^2 = 0$: equal weights are optimal-in-family, not merely safe; walk away. $\hat{\tau}^2 > 0$ under all three: the blocks truly differ. On our data acting on that signal failed ($k \leq 9$ blocks cannot support estimated weights, and the Hartung–Knapp intervals say why), but the detection was flawless, 5 of 5 folds and 3 of 3 estimators. On a future class with many more blocks ($k \geq 20$) or externally-known block identities, the tilt deserves a fresh trial: not fruitful to act on here, demonstrably correct as a detector.
4. The gate for shapes (Section 5.4). Run the stump fit and the two-stage bar. Its measured operating characteristics: 19 of 20 correct adoptions across feature subsets, one benign refusal, one prevented out-of-sample blow-up (class 3; the class-2 refusal rests on decisive in-sample

evidence), and refusals on classes 2 and 4 that cost the selected composite nothing on the hold-out. That is the conservative failure mode you want in production.

5. After deployment, monitor the level, not just the structure. Class 1's correlation matrix was rock-stable while its alpha halved, so a structural drift alarm alone would have missed the one failure that actually occurred (Figure 10). Track the trailing-year mean IC against the in-sample rolling distribution and alarm below its 5th percentile, treating structural drift as a separate, second alarm.

10.2 What we would explore with Ultramarin's resources

- Rolling windows and re-estimation, per future class. On our four classes rolling and scheduled re-estimation bought nothing, and the reason is diagnostic rather than universal. The composite has no weights to re-estimate, but it does have estimated structure: the clustering, the sign bits, the shape curves, and the gate decision. Our classes were structurally stable (hold-out correlation-matrix Frobenius drift between 0.16 and 0.42, unrelated to where performance fell), so refits had nothing to fix. Class 1's decline was alpha decay, which no re-estimation can recover, and the only rolling win anywhere (+0.0019, $t = 3.4$, on class 4) merely repaired the retired tilt's own frozen-weight damage. A future class could differ, so this must be re-examined per class, and the playbook's drift diagnostics are the instruments that decide. Structural drift says re-derive the clustering and signs, which is safe because the re-derived object is still an equal-weighted partition with nothing continuous to overfit. Level decay says retire or de-weight, not refit.
- Cross-class composition. Our mandate was within-class, but the four selected composites are themselves four correlated, noisy signals: exactly the structure this study composes. The natural next study runs the identical method one level up (dedup-then-equal-weight across class composites, τ^2 as the diagnostic), where block identities are known and economically meaningful.
- Real class identities. Every diagnostic here was run blind. Knowing the theme behind a class (momentum, quality, and so on) converts the per-class map from correlation into interpretation: whether uniform alpha decay of the class-1 kind matches that theme's crowding cycle, and whether the tail-concentrated shapes match its economic story. The shapes are lookup tables, and a portfolio manager can read them directly.
- More classes. The adaptive rule (diagnose, then act) has validated components but only four end-to-end trials. Twenty classes would let the rule itself be cross-validated, and would let the two ideas our data could detect but not reward (the τ^2 tilt, and feature-level GLS on multi-directional classes) be re-tried exactly where the diagnostics say they should work.
- Real cost models. Our tradeability axes (retention, turnover, half-life, slow-alpha share) are cost proxies. With Ultramarin's actual transaction-cost model, the weekly criterion becomes net-of-cost P&L, and the class-1 shape result, half the turnover at equal IC, converts into a directly bankable number.

10.3 Limitations

Several limitations should be kept in mind. Four classes is a small sample for every cross-class claim. The hold-out, though blind at delivery, was consulted repeatedly during the final review phase; every such use is labeled in the notebook, the final specification is frozen as of 2026-07-11,

and the next data delivery is the certification event, not another tuning opportunity. The weekly-criterion package is a post hoc combination of pre-specified components. The live composite's policy for day-level missing variants is untested, since the feature-subset tests remove variants permanently while complete-case preprocessing drops $\sim 12\%$ of rows on classes 2–3. Single-name costs, capacity, and portfolio construction are all outside scope. Finally, all classes share one market and one period, so cross-class pooled statistics inherit common shocks that our day-clustered inference only partially absorbs.

11 Conclusion

We were asked whether a cluster of correlated feature variants can be combined into something better than the rank-and-average benchmark. The disciplined answer is: not on raw IC. What can be beaten is the benchmark's *fragility*: its duplication bias on diffuse clusters, fixed by counting each idea once; its churn on classes with learnable shape structure, fixed on the evidence-backed class only by per-feature transforms that read as lookup tables; and its false promise that sophistication is free, documented by a catalogue of upgrades that failed rigorous tests before they could cost money. The selected composite is one sentence long, has no tuned parameters, survived a 2,180-day blind hold-out exactly as advertised (with the review-phase caveats of Section 10 owned in full), and comes packaged with the diagnostics a future user needs to know when its assumptions stop holding. We consider the negative results the most valuable deliverable: each one retired an upgrade before it could cost money.

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A Full hold-out scorecard

Table 4 reports the complete frozen scorecard on the blind hold-out. Hit rates, IC-IR, and portfolio IR tables are in the notebook (Part XI), and the full signal-life record is plotted in Appendix C.

Table 4: Blind hold-out, all frozen methods: mean daily rank IC and Newey–West t ($\sim 2,180$ days per class).

method	mean IC				NW t			
	c1	c2	c3	c4	c1	c2	c3	c4
benchmark (rank-and-average)	0.0118	0.0146	0.0133	0.0192	1.88	2.43	2.64	3.94
IC weighting	0.0119	0.0145	0.0135	0.0154	1.88	2.47	2.47	2.86
PCA PC1	0.0112	0.0123	0.0133	0.0063	1.78	2.18	2.66	1.64
PLS (tuned k)	0.0121	0.0157	0.0070	0.0184	1.94	2.62	1.76	3.40
uniqueness regression (tuned)	0.0119	0.0157	0.0084	0.0154	1.89	2.60	1.95	2.86
adaptive weights, raw variants	0.0117	0.0147	0.0133	0.0178	1.88	2.45	2.64	3.37
dedup + equal weights	0.0119	0.0170	0.0133	0.0194	1.91	2.71	2.64	3.69
dedup + adaptive weights	0.0119	0.0175	0.0133	0.0163	1.89	2.66	2.64	3.06
shaped composite	0.0154	0.0147	0.0027	0.0183	2.45	3.06	0.62	3.61
boosted trees	0.0149	0.0138	0.0049	0.0104	2.47	3.00	1.27	2.44

B Guide to the companion notebook

The full study is one executed, checkpointed Jupyter notebook (Ultramarin.ipynb, 200 cells). NOTEBOOK_SUMMARY.md gives per-section summaries and maps every section of this report to its notebook part (data and exploratory analysis are Part I, the framework Part II, the in-sample study Parts II–X, the hold-out Part XI, and the statistical checks Part XII). All randomness is seeded, heavy cells checkpoint their products to a cache file with a one-click resume path, and a full Restart-and-Run-All reproduces every number quoted here (last verified clean run: 2026-07-25, zero errors). Every figure in this report is generated from that checkpoint or the raw parquets by figures/make_v2_figures.py. The in-sample and hold-out panels are kept in separate parquet files and are never joined except in the two explicitly-labeled deployment simulations.

C Signal-life record on the hold-out

Figure 15 plots the full signal-life record on the blind hold-out (IC retention, factor autocorrelation, and IC earned by lag) for every frozen method on every class, complementing the scorecard of Appendix A.

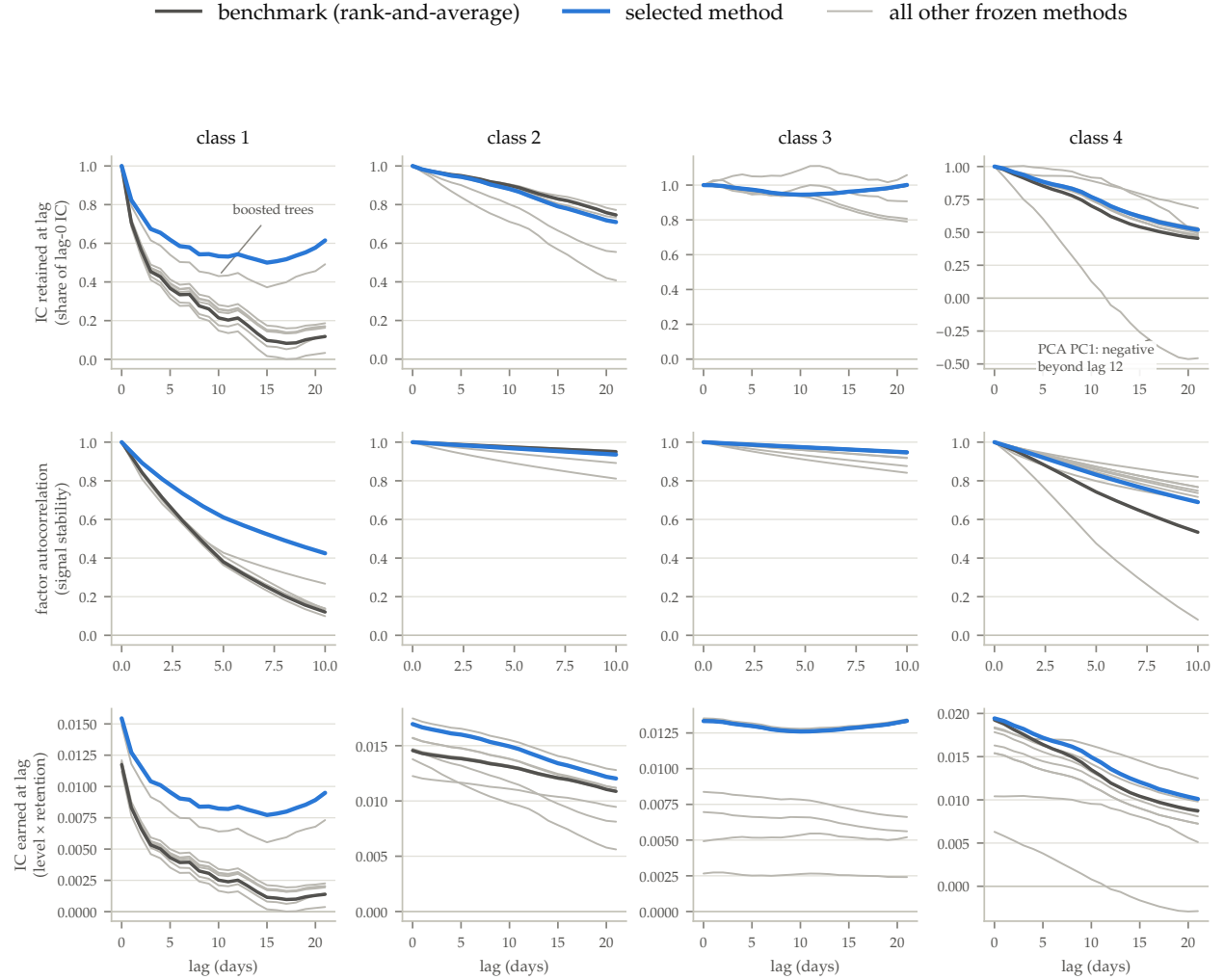


Figure 15: The full signal-life record on the blind hold-out, for every frozen method on every class. Top row: IC retained at each holding lag, as a share of the lag-0 IC. Middle row: the composite's own autocorrelation at each lag, a measure of how fast the signal itself changes. Bottom row: IC earned at each lag, the product of level and retention. Gray is the benchmark, blue the selected method for that class, and thin light-gray lines all other frozen methods. The class sets the clock: only class 1's shapes slow it down, and only class 4's PCA breaks it, going negative beyond lag 12.